

Programme: B E – Information Science & Engineering
Internal Assessment – I

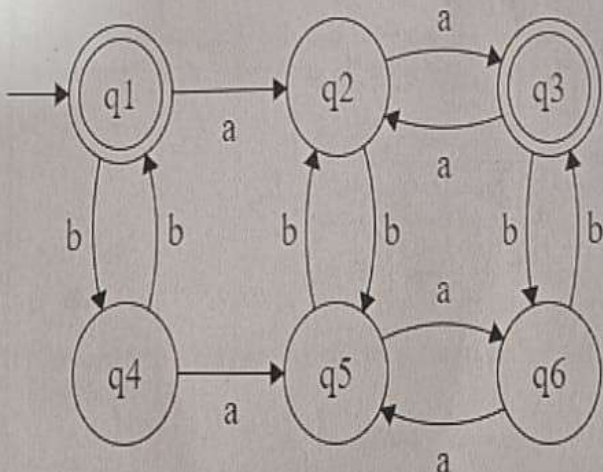
TERM : 10/09/2025-31/12/2025	COURSE NAME: Theory of Computation
DATE : 31/10/2025 TIME: 11:00am to 12:00pm	COURSE CODE : IS54
MAX MARKS: 30	PORTIONS : L1 - L 12

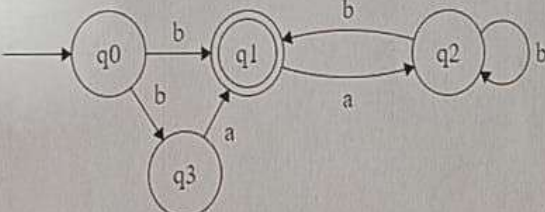
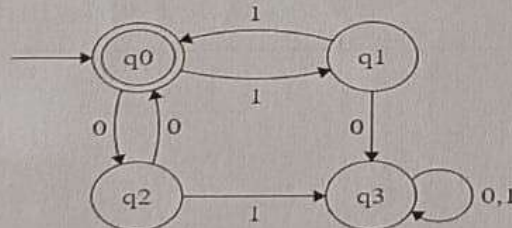


Mobile Phones are banned

Instructions to Candidates: **Answer any TWO full questions.**

Marks: **15x2=30**

Q. NO	Questions	Blooms Levels (L1 to L6)	CO	Marks
1.a	Define the following terms with examples. i) Alphabet ii) String iii) Epsilon iii) Kleens Star iv) Language. vii) Finite Automata	L1	CO1	7
b	Minimize DFA for the following state diagram. 	L3	CO2	8

2.a	Consider the following ϵ-NFA <table><tr><th>δ</th><th>ϵ</th><th>a</th><th>b</th></tr><tr><td>$\rightarrow p$</td><td>$\{r\}$</td><td>$\{q\}$</td><td>$\{p,r\}$</td></tr><tr><td>q</td><td>$\emptyset$</td><td>$\{p\}$</td><td>$\emptyset$</td></tr><tr><td>*r</td><td>$\{p,q\}$</td><td>$\{r\}$</td><td>$\{p\}$</td></tr></table> i) Compute the ϵ-closure of each state. ii) Convert the ϵ-NFA into its equivalent DFA by applying the Lazy construction method.	δ	ϵ	a	b	$\rightarrow p$	$\{r\}$	$\{q\}$	$\{p,r\}$	q	$\emptyset$	$\{p\}$	$\emptyset$	*r	$\{p,q\}$	$\{r\}$	$\{p\}$	L3	CO1	8
δ	ϵ	a	b																	
$\rightarrow p$	$\{r\}$	$\{q\}$	$\{p,r\}$																	
q	$\emptyset$	$\{p\}$	$\emptyset$																	
*r	$\{p,q\}$	$\{r\}$	$\{p\}$																	
b	Define regular expressions. Convert the following regular expressions to NFA with ϵ-Transitions. i) $(00)^*11(0+1)^*$ ii) $a^*+b^*+c^*$	L3	CO2	7																
3.a	Define DFA. Construct DFA for the following languages: i) $L=\{ W \in \{0,1\}^* \mid W \text{ is of even length and begins with } 00\}$ ii) $L= \{W \in \{a,b\}^* \mid \text{No two consecutive characters are same}\}$ iii) $L= \{W \in \{0,1\}^* \mid W \text{ that not ends with } 00 \text{ or } 11\}$	L3	CO1	7																
b	Discuss any two applications of regular expressions. Obtain regular expression (RE) from given finite automata by using state elimination method i.  ii. 	L3	CO2	8																

* L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create

Programme : B E – Information Science & Engineering
Internal Assessment – II

TERM : 10/9/25 to 31/12/25	COURSE NAME: Theory of Computation
DATE : 26/12/25 TIME: 11-12pm	COURSE CODE : IS54
MAX MARKS: 30	PORTIONS : L18-L42



Instructions to Candidates: Answer any TWO full questions.

Marks: 15x2=30

Q. NO	Questions	Blooms Levels	CO	Marks
1.a	Obtain CFGs for the following languages: i) Consisting of all balanced parentheses. ii) $L = \{w \mid n_a(w) = n_b(w)\}$	L3	CO3	5
b	Write transition functions for the PDA to accept language $L = \{a^n b^n \mid n \geq 1\}$. Trace it for the string aabbb.	L3	CO4	5
c	Prove that every language accepted by a multitape TM is recursively enumerable.	L2	CO5	5
2.a	Define Nullable symbol. Eliminate ϵ -production from the given grammar: $S \rightarrow ASA \mid aBbA$ $A \rightarrow B \mid BAa$ $B \rightarrow b \mid \epsilon$	L3	CO3	5
b	Define NPDA. Design a PDA for the given CFG: $S \rightarrow aABC$ $A \rightarrow aB \mid a$ $B \rightarrow bA \mid b$ $C \rightarrow a$	L3	CO4	5
c	With the help of an example, define NP Hard and NP complete.	L2	CO5	5
3.a	Perform LMD for the string aaabbabbbba using the below grammar: $S \rightarrow aB \mid bA$ $A \rightarrow aS \mid bAA \mid a$ $B \rightarrow bS \mid aBB \mid b$	L3	CO3	5
b	Convert the following PDA to equivalent CFG: $\delta(q_0, a, Z) = (q_0, AZ)$ $\delta(q_0, b, A) = (q_0, AA)$ $\delta(q_0, a, A) = (q_1, \epsilon)$	L3	CO4	5
c	Write transitions of the TM for the minus operation when the condition is $m > n$.	L3	CO5	5

* L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create



SEMESTER END / BACKLOG SUBJECT EXAMINATIONS - JANUARY 2026

Program : **B.E. - Information Science and Engineering**
Course Name : **Theory of Computation**
Course Code : **21 / 22 / 23 / IS54**

Semester : **V**
Max. Marks : **100**
Duration : **3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

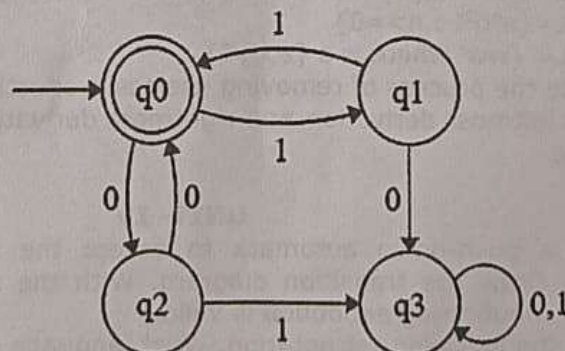
- Discuss the central concepts of automata theory with the help of examples. CO1 (04)
 - Draw a DFA to accept the following language: CO1 (08)
 - $L = \{w(ab+ba) \mid w \in (a,b)^*\}$
 - Strings of a's and b's except those having the substring "aab"
 - Strings of 0's and 1's having three consecutive 0's
 - strings of a's and b's having even numbers of a's and odd numbers b's.
 - With the help of an example, discuss the various components of Deterministic finite automata (DFA). CO1 (08)
- With an example illustrate the process of epsilon closures computation in automata. CO1 (04)
 - Design a DFA for the NFA language $L = \{w \mid w \text{ ends in } 01\}$ using Lazy evaluation. CO1 (08)
 - Differentiate between DFA, NFA and ϵ - NFA. CO1 (08)

UNIT - II

- Construct ϵ -NFA for the regular expression $(a+b)^*ab$. CO2 (05)
 - Minimize the following DFA using table filling algorithm. CO2 (10)

δ	0	1
$\rightarrow A$	B	A
B	A	C
C	D	B
*D	D	A
E	D	F
F	G	E
G	F	G
H	G	D

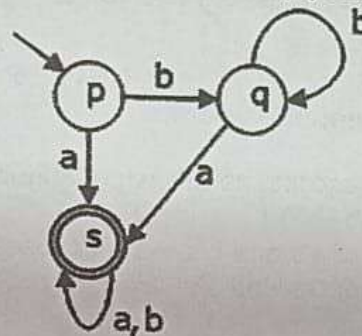
- Convert the following FA to a regular expression using state elimination method- CO2 (05)



4. a) What is regular expression? Write regular expressions for the following languages: CO2 (06)
- $L = \{a^n b^m \mid n \geq 4, m \leq 3\}$
 - $L = \{a^m b 2^n \mid m, n \geq 0\}$
- b) Apply $R_{ij}^{(k)}$ method to find the equivalent regular expression for the given finite automata- CO2 (09)

	0	1
$\rightarrow q_0$	q1	q0
$*q_1$	q1	q1

- c) What are indistinguishable states? Are p and q indistinguishable states in the given FA for the input string 'bba'? Justify your answer. CO2 (05)



UNIT - III

5. a) Convert the given grammar into Chomsky normal form by eliminating ϵ production, unit production and useless production. CO3 (10)
- $$\begin{aligned}
 S &\rightarrow AaBa \mid ASBb \\
 A &\rightarrow aAbB \mid a \mid B \\
 B &\rightarrow SbS \mid AC \mid bb \mid \epsilon \\
 C &\rightarrow aC \mid c \\
 D &\rightarrow d
 \end{aligned}$$
- b) With the help of an example, explain the process of constructing a parse tree. CO3 (05)
- c) Is the following grammar ambiguous? CO3 (05)
- $$\begin{aligned}
 S &\rightarrow aB \mid bA \\
 A &\rightarrow aS \mid bAA \mid a \\
 B &\rightarrow bS \mid aBB \mid b
 \end{aligned}$$

6. a) Design context free grammars for the following language. CO3 (10)
- $L = \{w \mid n_a(w) = n_b(w), w \in (a+b)^*\}$
 - $L = \{a^n b^m \mid n \geq 0, m > n\}$
 - $L = \{w : n_a(w) \bmod 2 = 0 \text{ where } w \in (a,b)^*\}$
 - $L = \{a^n b^{2n} : n \geq 0\}$
 - $L = \{ww^r \text{ where } w \in \{a,b\}^*\}$
- b) Illustrate the process of removing useless production from a grammar. CO3 (05)
- c) Discuss leftmost derivation and rightmost derivation with the help of an example. CO3 (05)

UNIT- IV

7. a) Design a push-down automata to accept the language $L = \{a^n b^{3n}, n \geq 1\}$. Draw the transition diagram. With the sequence of IDs show that the input string aabbbbbbb is valid. CO4 (10)
- b) Explain the following set notation. What language does it indicate? CO4 (05)
- $$L = \{w \mid (q_0, w, Z_0) \vdash^* (q, \epsilon, \epsilon)\}$$

c) Are the following transitions for NPDA? Give justification for your answer. CO4 (05)

- i) $\delta(q_0, (, Z_0) = (q_1, Z_0)$
- ii) $\delta(q_0, [, Z_0) = (q_1, [Z_0)$
- iii) $\delta(q_1, (, () = (q_1, (()$
- iv) $\delta(q_1, (, [) = (q_1, ([)$
- v) $\delta(q_1, [, () = (q_1, [()$
- vi) $\delta(q_1, [, [) = (q_1, [[)$
- vii) $\delta(q_1,), () = (q_1, \epsilon)$
- viii) $\delta(q_1,], [) = (q_1, \epsilon)$
- ix) $\delta(q_1, \epsilon, Z_0) = (q_0, Z_0)$

8. a) What type of strings are accepted by the following PDA? Describe each of the transitions. CO4 (06)

- i) $\delta(q_0, a, Z_0) = (q_0, aZ_0)$
- ii) $\delta(q_0, a, a) = (q_0, aa)$
- iii) $\delta(q_0, b, a) = (q_1, \epsilon)$
- iv) $\delta(q_1, b, a) = (q_1, \epsilon)$
- v) $\delta(q_1, \epsilon, Z_0) = (q_f, Z_0)$

b) Design a push-down automata to accept the language $L = \{N_a(w) = N_b(w) \mid w \in (a+b)^*\}$. Is it DPDA? Justify your answer. CO4 (08)

c) Construct a PDA for the following CFG-
 $S \rightarrow aA B C, A \rightarrow a B \mid a, B \rightarrow b A \mid b, C \rightarrow a.$ CO4 (06)

UNIT - V

9. a) Prove that every language accepted by a Multitape TM is recursively enumerable. CO5 (10)

b) With the help of example, explain NP-Hard and NP-complete. CO5 (05)

c) Illustrate storage in the state in turing machine. CO5 (05)

10. a) Design a Turing machine to accept the language $L = \{0^n 1^n \mid n \geq 1\}$. Prove that string the "000111" is valid or not. CO5 (10)

b) Illustrate the working of a turning machine using block diagram. CO5 (05)

c) Explain the halting problem in a turning machine. CO5 (05)
